Physical Chemistry Cume: Phase Changes

1. Define the critical temperature and critical pressure of a substance. For instance, what does it mean that the critical temperature of CO_2 is 31.0 °C (305.2 K) and its critical pressure is 73.8 bar (7.38 MPa). (40 p)

2. How can one determine the critical temperature and pressure experimentally? (20 p)

3. Give an idea how big a pressure is 73.8 bar. It is very crudely, i.e., within 30 %
(i) a tenth of the atmospheric pressure
(ii) about the same as the atmospheric pressure
(iii) ten times the atmospheric pressure
(iv) hundred times the atmospheric pressure,
(v) thousand times the atmospheric pressure
(select one). (10 p)

4. How is the SI unit of pressure, the Pascal (Pa) defined? Express it with the basic SI units kg, m and s. (20 p)

5. Give an idea of the magnitude of the Pascal. How much is the standard atmospheric pressure in Pa units (approximately)? (10 p)